THE SIX-COLOR CASE OF AN ERDŐS–GYÁRFÁS BALANCED-COLORING PROBLEM

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Abstract. We prove the six-color case of a conjecture of Erdős and Gyárfás: every six-coloring of the edges of K_{37} has seven vertices whose induced edges omit at least one color. For a fixed color, the counterexample hypothesis gives a graph in which every seven vertices span at most sixteen edges. We combine the Kang–Pikhurko extremal theorem with minimum-degree decompositions to prove clique lemmas on 18, 19, 24, and 25 vertices. These lemmas give a 97-edge lower bound for the relevant 31-vertex graphs and then contradict the edge count of a least color on 37 vertices. The proof is non-computational. This is a fixed-parameter result and does not settle the conjecture for general r.

Preprint. This manuscript has not completed external mathematical review.

AI-use disclosure. OpenAI GPT-5.6 Sol was used extensively during proof search, drafting, and internal checking. OpenAI GPT-5 (Codex) was used for later source review and release checks. xAI Grok 4.5 was used to critique a draft. This manuscript has not completed external mathematical review.

1. Introduction

Erdős and Gyárfás asked whether the following assertion holds for every integer $r \geq 3$: whenever the edges of K_{r^2+1} are colored with r colors, some $r+1$ vertices induce a complete graph on which at least one color is absent [3]. They proved the cases $r = 3$ and $r = 4$, observed that the analogous assertion fails for $r = 2$, and gave related constructions on r^2 vertices. The question is also recorded as Erdős Problem 617 [1].

We prove the assertion for six colors.

Theorem 1.1. *For every map $\chi : E(K_{37}) \to [6]$, there is a set $S \subseteq V(K_{37})$ with $|S| = 7$ such that*

$$\chi(E(K_{37}[S])) \neq [6].$$

Theorem 1.1 concerns only $r = 6$. No assertion for $r \geq 7$ is proved here.

For one color in a hypothetical counterexample, let G be the graph formed by the edges of that color. Every seven vertices span at most sixteen edges of G, since each of the other five colors must occur. We call this the local cap. The argument has three stages. First, we prove that certain admissible graphs on 18, 19, 24, and 25 vertices contain K_6. Second, we peel three such

Date: 18 July 2026.

2020 *Mathematics Subject Classification.* 05C15, 05C35, 05D10.

Key words and phrases. edge coloring, balanced coloring, extremal graph theory, Ramsey theory.

cliques from a 31-vertex graph and reach a triangle-free 13-vertex complement that violates the local cap. Finally, a least color on 37 vertices reduces by averaging to that 31-vertex obstruction.

The only external graph bounds used below are the Kang–Pikhurko theorem [4] and Brooks's theorem [2]. All finite case splits in the proof are given explicitly.

2. Preliminaries

All graphs are finite and simple. For a graph F, we write $e(F)$ for its number of edges, $\alpha(F)$ for its independence number, $\tau(F)$ for its minimum vertex-cover number, and $\delta(F)$ and $\Delta(F)$ for its minimum and maximum degrees. The complement of F is $\overline{F}$, and $F[W]$ denotes the subgraph induced by $W \subseteq V(F)$. The Turán graph with q parts on n vertices is $T_q(n)$, and $t_q(n) = e(T_q(n))$.

Definition 2.1. A graph F is *admissible* if

$$e(F[S]) \leq 16 \qquad \text{for every } S \in \binom{V(F)}{7}.$$

If F is admissible and $L = \overline{F}$, then every seven vertices span at least five edges of L. Admissibility is inherited by induced subgraphs.

We use the following part of the Kang–Pikhurko theorem, together with its equality characterization [4, Theorems 1 and 4 and Lemma 5].

Theorem 2.2 (Kang–Pikhurko)**.** *Let $q \geq 2$, let $n \geq q + 3$, and suppose $q \leq (n-1)/2$. If L is a K_{q+1}-free graph on n vertices that is not q-partite, then*

$$e(L) \leq t_q(n) - \left\lfloor \frac{n}{q} \right\rfloor + 1. \tag{1}$$

The equality graphs are those in the construction characterized by Kang and Pikhurko.

We use Brooks's theorem in the following standard form: a connected graph of maximum degree D has a proper D-coloring unless it is complete or an odd cycle [2].

One fixed equality exclusion will be used to reach the lower endpoint in the 18-vertex lemma.

Lemma 2.3 (The 18-vertex endpoint)**.** *Let R be an admissible graph on 18 vertices with $\alpha(R) \leq 3$. If $\overline{R}$ is not three-partite, then*

$$e(R) \geq 51.$$

Proof. The graph $L = \overline{R}$ is K_4-free and not three-partite. Since $t_3(18) = 108$, Theorem 2.2 first gives

$$e(L) \leq 108 - 6 + 1 = 103, \qquad e(R) \geq 50.$$

Suppose equality holds. For $q = 3$ and $n = 18$, Lemma 5 of Kang and Pikhurko says that the nondecreasing old part sizes n_1, n_2, n_3 sum to seventeen and satisfy

$$n_1 \geq 2, \qquad n_2 \leq n_1 + 1, \qquad n_3 \leq n_1 + 2.$$

If $n_1 \leq 4$, their sum is at most $4+5+6=15$, while if $n_1 \geq 6$, their sum is at least eighteen. Thus $n_1 = 5$, and the only old part-size sequences are $(5,6,6)$ and $(5,5,7)$. In the complement R, every old part is a clique. The second sequence therefore contains a K_7, contrary to admissibility.

For $(5,6,6)$, let N_s and N_t be the special and witness parts in the equality construction, let x be its exceptional vertex, let $\varnothing \neq A \subsetneq N_s$ be its distinguished subset, and let $y \in N_t$ be its witness vertex. In L, the vertex x is adjacent to $A \cup \{y\}$ and to every old part other than N_s, N_t, while the edges between y and A are deleted. Hence, in R, the vertex x is adjacent to $N_s \setminus A$ and $N_t \setminus \{y\}$, and y is adjacent to A.

The parts N_s, N_t have sizes five and six, in either order. If the special part has size five, the witness K_6 together with the exceptional vertex spans

$$\binom{6}{2} + 5 = 20$$

edges of R. If the special part has size six, let a, with $1 \leq a \leq 5$, be the size of its distinguished subset. The special K_6, together with either the exceptional vertex or the witness vertex, spans

$$\binom{6}{2} + \max(a, 6-a) \geq 18$$

edges. Each case exceeds the local cap of sixteen. Equality is impossible, so $e(R) \geq 51$. $\square$

The next accounting observation will be used repeatedly.

Lemma 2.4 (Neighborhood accounting)**.** *Let F be a graph, let v be a minimum-degree vertex of degree d, and put*

$$N = N_F(v), \qquad U = V(F) \setminus (N \cup \{v\}),$$
$$X = e_F(N, U), \qquad Y = e(F[N]), \qquad A = X + Y.$$

Then

$$X + 2Y \geq d(d-1), \qquad A \geq \binom{d}{2}. \tag{2}$$

Consequently,

$$e(F[U]) \leq e(F) - d - \binom{d}{2}. \tag{3}$$

If $\alpha(F) \leq s+1$, then $\alpha(F[U]) \leq s$.

Proof. Each vertex of N has at least $d-1$ incident edges after its edge to v is removed. Summing over N counts an N–U edge once and an edge of $F[N]$ twice, which proves the first inequality. Since $Y \leq \binom{d}{2}$,

$$A = (X + 2Y) - Y \geq d(d-1) - \binom{d}{2} = \binom{d}{2}.$$

The edge decomposition

$$e(F) = d + A + e(F[U])$$

gives (3). An independent $(s+1)$-set in $F[U]$ together with v would be an independent $(s+2)$-set in F. $\square$

We will use the following elementary averaging fact.

Lemma 2.5 (Induced-subgraph averaging)**.** *Let F have n vertices and E edges, and let $2 \le q \le n$. Some induced q-vertex subgraph of F has at most*

$$E\frac{\binom{q}{2}}{\binom{n}{2}}$$

edges.

Proof. Choose a q-set uniformly at random. Each edge of F belongs to the chosen set with probability $\binom{q}{2}/\binom{n}{2}$, so the expected number of induced edges is the displayed quantity. At least one choice has no more edges than the expectation. □

We also record the effect of removing a K_6.

Lemma 2.6 (Clique peeling)**.** *Let F be admissible, let $B \subseteq V(F)$ span a K_6, and suppose $\alpha(F) \le s \le 5$. Then*

$$\alpha(F - B) \le s - 1, \qquad e(F - B) \le e(F) - 15.$$

Proof. Every vertex outside B has at most one neighbor in B, since two such neighbors together with B would give at least seventeen edges on seven vertices. If $I \subseteq V(F) \setminus B$ were an independent s-set, the edges from I to B would meet at most $s < 6$ vertices of B. A vertex of B outside their endpoints would extend I to an independent $(s+1)$-set. The edge bound follows by deleting the fifteen edges of B. □

The terminal argument below will appear at orders 13 and 14.

Lemma 2.7 (Terminal complement)**.** *Let L be triangle-free with $\alpha(L) \le 5$, and suppose every seven vertices span at least five edges. There is no degree-five vertex of L having at least seven nonneighbors.*

Proof. Suppose w is such a vertex. Put $A = N_L(w)$ and $B = V(L) \setminus (A \cup \{w\})$, so $|A| = 5$ and $|B| \ge 7$. The set A is independent. For $b \in B$, write

$$t_b = |N_L(b) \cap A|.$$

If $b, c \in B$ satisfy $t_b, t_c \le 2$, then the seven-set $A \cup \{b, c\}$ spans

$$t_b + t_c + \mathbf{1}_{bc \in E(L)}$$

edges. The local lower bound forces $t_b = t_c = 2$ and $bc \in E(L)$. Thus the vertices with $t_b \le 2$ form a clique, and triangle-freeness permits at most two of them.

At least five vertices of B therefore satisfy $t_b \ge 3$. They are pairwise nonadjacent: if two were adjacent, their neighborhoods in the independent set A would be disjoint, although their sizes sum to at least six. Five such vertices together with w form an independent six-set, contrary to $\alpha(L) \le 5$. □

3. The 18-vertex clique lemma

We now prove the full statement denoted by P_3 in the reduction.

Proposition 3.1 (P_3)**.** *If R is admissible on 18 vertices, $\alpha(R) \leq 3$, and $e(R) \leq 55$, then R contains a K_6.*

Proof. If $\overline{R}$ is three-partite, one of its three parts has size at least six and gives a K_6 in R. Otherwise Lemma 2.3 gives $e(R) \geq 51$.

Assume that R has no K_6. Choose a minimum-degree vertex v, write $d = d_R(v)$, and let U be its $17 - d$ nonneighbors. Since $\alpha(R[U]) \leq 2$, the graph

$$L = \overline{R[U]}$$

is triangle-free. The absence of a K_6 gives $\alpha(L) \leq 5$. Every neighborhood in a triangle-free graph is independent, so

$$\Delta(L) \leq 5. \tag{4}$$

Lemma 2.4 and (4) give the following bounds.

d	$\lvert U\rvert$	lower bound on $e(R[U])$	lower bound on $e(R)$
0	17	94	94
1	16	80	81
2	15	68	71
3	14	56	62
4	13	46	56

Thus $d \geq 5$. Since $e(R) \leq 55$, the average degree gives $d \leq 6$.

Suppose first that $d = 5$. Now $|U| = 12$, and (4) gives $e(R[U]) \geq 36$. Write

$$e(R) = 50 + \ell, \qquad e_R(N, U) + e(R[N]) = 10 + a,$$
$$e(R[U]) = 36 + k.$$

Here $1 \leq \ell \leq 5$, and exact edge accounting gives

$$a + k = \ell - 1, \qquad a, k \geq 0. \tag{5}$$

Let $M = \overline{R[N]}$, put $m = e(M)$, and for $x \in N$ write $x_x = d_R(x, U)$. Minimum degree gives

$$x_x \geq d_M(x). \tag{6}$$

The sum of the cross-degrees is

$$\sum_{x \in N} x_x = (10 + a) - (10 - m) = a + m.$$

Summing (6) gives $a + m \geq 2m$, so

$$1 \leq m \leq a \leq 4. \tag{7}$$

The lower bound $m \geq 1$ follows because $m = 0$ would make $N[v]$ a K_6.

Since $\alpha(L) \leq 5$,

$$\tau(L) = 12 - \alpha(L) \geq 7.$$

If $pq \in E(M)$, then

$$N_R(p, U) \cup N_R(q, U) \tag{8}$$

is a vertex cover of L. Indeed, an uncovered edge of L together with p, q would be an independent four-set in R.

There is an edge $pq \in E(M)$ for which $x_p + x_q \le 6$. If $m \le 2$, this follows from

$$\sum_{x \in N} x_x = a + m \le 6.$$

If $m = 3$, the total is at most seven, and every edge of a three-edge simple graph has a nonisolated vertex outside its endpoints. That outside vertex has positive cross-degree by (6). If $m = 4$, the total is eight, while a four-edge simple graph has at least four nonisolated vertices. At least two positive cross-degrees therefore lie outside the endpoints of any chosen edge. In every case (8) has size at most six, contrary to $\tau(L) \ge 7$.

It remains that $d = 6$. Now $|U| = 11$. Put

$$X = e_R(N, U), \qquad Y = e(R[N]).$$

The local cap on $N[v]$ gives $Y \le 10$, while minimum degree gives

$$X + 2Y \ge 30, \qquad X + Y \ge 20.$$

The graph $L = \overline{R[U]}$ is triangle-free and nonbipartite. Applying Theorem 2.2 with $q = 2$ and $n = 11$ gives

$$e(L) \le t_2(11) - \left\lfloor \frac{11}{2} \right\rfloor + 1 = 30 - 5 + 1 = 26.$$

Consequently,

$$e(R[U]) \ge \binom{11}{2} - 26 = 29$$

and

$$e(R) \ge 6 + 20 + 29 = 55.$$

All inequalities are equalities. Thus

$$e(R) = 55, \qquad Y = 10, \qquad X = 10, \qquad e(L) = 26.$$

Let $M = \overline{R[N]}$. Then M has five edges. For each $x \in N$, minimum degree gives $d_R(x, U) \ge d_M(x)$; equality of the sums gives

$$d_R(x, U) = d_M(x) \qquad (x \in N). \tag{9}$$

We also have $\tau(L) = 11 - \alpha(L) \ge 6$. As above, every edge $pq \in E(M)$ makes the two U-neighborhoods a vertex cover of L. Hence

$$d_M(p) + d_M(q) \ge 6 \qquad (pq \in E(M)). \tag{10}$$

We classify a five-edge simple graph M on six vertices satisfying (10). If M has a leaf, its neighbor has degree five, and all five edges form $K_{1,5}$. If M has no leaf, let t be the number of nonisolated vertices. The ten edge-ends give $4 \le t \le 5$. For $t = 4$, the graph is K_4 minus one edge, and an edge joining degrees two and three violates (10). For $t = 5$, every nonisolated vertex has degree two, so $M = C_5 \,\dot\cup\, K_1$, which also violates (10). Therefore $M = K_{1,5}$. Its five leaves form a K_5 in $R[N]$, and adjoining v gives a K_6, the final contradiction. $\square$

4. The 19-vertex clique lemma

Proposition 4.1. *If Q is admissible on 19 vertices, $\alpha(Q) \le 3$, and $e(Q) \le 66$, then Q contains a K_6.*

Proof. Suppose not. Choose a minimum-degree vertex v, put $d = d_Q(v)$, and let U be its $18-d$ nonneighbors. The graph

$$L = \overline{Q[U]}$$

is triangle-free because $\alpha(Q[U]) \le 2$. Since $Q[U]$ has no K_6, $\alpha(L) \le 5$, and hence $\Delta(L) \le 5$.

Write $X = e_Q(N(v), U)$ and $Y = e(Q[N(v)])$. Lemma 2.4 gives

$$X + 2Y \ge d(d-1), \qquad X + Y \ge \binom{d}{2}.$$

For $d = 0, 1, 2, 3$, the resulting lower bounds on $e(Q)$ are

$$108, \qquad 95, \qquad 83, \qquad 74,$$

respectively. The average degree therefore leaves $d \in \{4, 5, 6\}$.

If $d = 4$, then $|U| = 14$ and

$$e(Q[U]) \ge \binom{14}{2} - 35 = 56.$$

All bounds are equalities. In particular, $e(Q) = 66$, $Y = 6$, and $X = 0$. Thus $N[v]$ is an isolated K_5, and L is a 5-regular triangle-free graph on 14 vertices. Every seven vertices span at least five edges of L. Any vertex of L has degree five and eight nonneighbors, contrary to Lemma 2.7.

Suppose $d = 5$. Let

$$M = \overline{Q[N(v)]}, \qquad m = e(M),$$

and write

$$x_a = d_Q(a, U) = d_M(a) + z_a, \qquad z_a \ge 0, \qquad z = \sum_a z_a.$$

The 13-vertex graph L satisfies

$$e(L) \le 32, \qquad \tau(L) = 13 - \alpha(L) \ge 8.$$

Exact accounting gives

$$e(Q) = 5 + (10 - m) + (2m + z) + (78 - e(L)) = 93 + m + z - e(L) \le 66.$$

Since $e(L) \le 32$,

$$66 \ge 93 + m + z - e(L) \ge 61 + m + z,$$

so $m + z \le 5$. If $m = 0$, then $N[v]$ is a K_6. Otherwise choose $aa' \in E(M)$. The set

$$N_Q(a, U) \cup N_Q(a', U)$$

is a vertex cover of L, for an uncovered edge would extend a, a' to an independent four-set in Q. In the next degree sum the edge aa' is counted twice, while each of the other $m-1$ edges is counted at most once. Thus the cover has size at most

$$d_M(a) + d_M(a') + z_a + z_{a'} \le m + 1 + z \le 6,$$

contrary to $\tau(L) \geq 8$.

It remains that $d = 6$. Now $|U| = 12$. Retain the notation M, m, x_a, z_a and put $Z = \sum_a z_a$. The local cap on $N[v]$ gives $m \geq 5$. Since $e(L) \leq 30$, exact accounting gives

$$e(Q) = 6 + (15 - m) + (2m + Z) + (66 - e(L)) = 87 + m + Z - e(L) \leq 66.$$

Since $e(L) \leq 30$,

$$66 \geq 87 + m + Z - e(L) \geq 57 + m + Z,$$

and hence

$$Z \leq 9 - m. \tag{11}$$

We also have $\tau(L) \geq 12 - 5 = 7$. For every edge $aa' \in E(M)$, the two U-neighborhoods cover L, so

$$x_a + x_{a'} \geq 7. \tag{12}$$

We first suppose that M is triangle-free. If $\Delta(M) = 5$, then $M = K_{1,5}$, and its leaves form a K_5 in $Q[N(v)]$ that extends with v to a K_6. We may therefore assume $\Delta(M) \leq 4$. For every edge aa' of a triangle-free graph on six vertices,

$$d_M(a) + d_M(a') \leq 6.$$

Indeed, $N_M(a) \setminus \{a'\}$ and $N_M(a') \setminus \{a\}$ are disjoint subsets of the other four vertices. Summing (12) over all edges of M gives

$$7m \leq \sum_a d_M(a) x_a = \sum_a d_M(a)^2 + \sum_a d_M(a) z_a. \tag{13}$$

If $m \geq 8$, the first sum is at most $6m$, and the second is at most $4(9 - m)$. Their sum is smaller than $7m$. If $m = 7$ and $\Delta(M) \leq 3$, the right side is at most $42 + 6 < 49$. If $m = 7$ and $\Delta(M) = 4$, the four neighbors of a degree-four vertex are independent, and the three remaining edges join the sixth vertex to three of them. The degree sequence is $(4, 3, 2, 2, 2, 1)$, so the right side is at most $38 + 8 < 49$.

Let $m = 6$. If $\Delta(M) \leq 3$, then

$$\sum_a d_M(a)^2 \leq 30. \tag{14}$$

To see this, the only degree partitions of sum twelve and maximum part three with larger square-sum are $(3, 3, 3, 3, 0, 0)$ and $(3, 3, 3, 2, 1, 0)$. The first would induce K_4 on its four nonisolated vertices. In the second, deleting the isolated vertex leaves a triangle-free graph with six edges on five vertices. Choose a degree-three vertex u. Its three neighbors are independent, and the fifth vertex is the unique nonneighbor of u. Since only three edges remain after the edges incident with u are removed, all three must join that fifth vertex to the three neighbors of u. The resulting degree sequence is $(3, 3, 2, 2, 2)$, not $(3, 3, 3, 2, 1)$. Thus the right side of (13) is at most $30 + 9 < 42$. If $m = 6$ and $\Delta(M) = 4$, the degree sequence is $(4, 2, 2, 2, 1, 1)$. Equality in (13) would require all three units of Z to lie on the degree-four vertex. One of the two edges not incident with that vertex would then have endpoint weight $2 + 2 < 7$.

Let $m = 5$. If $\Delta(M) \le 2$, the right side of (13) is at most $20 + 8 < 35$. If $\Delta(M) = 3$, the sum can reach 35 only if the degree square-sum is at least 23. The only possible partitions are

$$(3,3,3,1,0,0), \qquad (3,3,2,2,0,0), \qquad (3,3,2,1,1,0).$$

The first is not graphical: three degree-three vertices on four nonisolated vertices force the fourth degree to be three. The second is K_4 minus an edge and contains a triangle. In the third, if the two degree-three vertices are adjacent, their two remaining neighbor sets among three vertices intersect and form a triangle. If they are nonadjacent, both are adjacent to all three remaining nonisolated vertices, which makes the two degree-one entries impossible. Finally, if $\Delta(M) = 4$, then M is $K_{1,4}$ together with an edge from the sixth vertex to one leaf. That last edge forces all four units of Z onto its endpoints. A different center–leaf edge then has endpoint weight $4 + 1 < 7$.

This excludes every triangle-free M except the star already handled. Suppose now that M contains a triangle T but no K_4, since a K_4 in M is an independent four-set in Q. Let W be the other three vertices, and put

$$\gamma = e_M(T, W), \qquad q = e(M[W]).$$

The three vertices of T are pairwise nonadjacent in Q. Their U-neighborhoods cover all twelve vertices of U, since an uncovered vertex would extend T to an independent four-set. Hence

$$12 \le \sum_{a \in T} x_a = 6 + \gamma + \sum_{a \in T} z_a \le 6 + \gamma + Z \le 12 - q. \tag{15}$$

Every inequality is an equality. Thus $q = 0$, all slack lies on T, and the three triangle weights sum to twelve.

Since $m = 3 + \gamma \ge 5$, at least one cross-edge exists. Each vertex of W has at most two neighbors in T, because M has no K_4, and it has zero slack. A triangle vertex incident with a cross-edge therefore has weight at least five by (12). Cross-edges cannot meet all three vertices of T, since their weights would sum to at least fifteen. They cannot meet only one, say a: every cross-neighbor would have weight one, forcing $x_a \ge 6$, while the opposite triangle edge would have endpoint-weight sum at most six.

Thus the cross-edges meet exactly two triangle vertices, say a, b. Every cross-neighbor meets both. Indeed, if a cross-neighbor met only a, then $x_a \ge 6$; since b is also incident with a cross-edge, $x_b \ge 5$, while $x_c \ge d_M(c) = 2$, contradicting the triangle weight sum twelve. Every cross-neighbor consequently has weight two. The edge constraints and the weight sum force

$$(x_a, x_b, x_c) = (5, 5, 2).$$

The three U-neighborhoods partition U into sets A, B, P of sizes $5, 5, 2$. Since $ac \in E(M)$, the set $A \cup P$ covers L. Hence B is independent in L, and $B \cup \{b\}$ is a K_6 in Q, the final contradiction. $\square$

5. The 25- and 24-vertex clique lemmas

Proposition 5.1. *If H is admissible on 25 vertices, $\alpha(H) \le 4$, and $e(H) \le 81$, then H contains a K_6.*

Proof. Suppose not, and choose a minimum-degree vertex v of degree $d \le 6$. If $d \le 5$, its $24 - d$ nonneighbors induce a graph with independence number at most three and, by Lemma 2.4, at most

$$81 - d - \binom{d}{2}$$

edges. Lemma 2.5, with $q = 19$, gives the corresponding expected edge bound below.

d	number of nonneighbors	expected edge bound
0	24	4617/92
1	23	13680/253
2	22	4446/77
3	21	855/14
4	20	639/10
5	19	66

Every entry is at most 66, so some induced 19-vertex graph satisfies the hypotheses of Proposition 4.1 and contains a K_6.

It remains that $d = 6$. Let $N = N_H(v)$, let R be the 18 nonneighbors of v, and put

$$m = 15 - e(H[N]).$$

The local cap on $N[v]$ gives $m \ge 5$. Each vertex of N needs at least its missing degree in $H[N]$ as cross-degree into R, so at least $2m$ edges join N to R. Therefore

$$e(R) \le 81 - 6 - (15 - m) - 2m = 60 - m \le 55.$$

Also $\alpha(R) \le 3$. Proposition 3.1 supplies a K_6. □

The next statement is the input denoted by P_4.

Proposition 5.2 (P_4)**.** *If P is admissible on 24 vertices, $\alpha(P) \le 4$, and $e(P) \le 70$, then P contains a K_6.*

Proof. Suppose not. A minimum-degree vertex v has degree $d \le 5$. If $d \le 4$, Lemmas 2.4 and 2.5, with $q = 19$, give the following expected edge bounds among the $23 - d$ nonneighbors.

d	number of nonneighbors	expected edge bound
0	23	11970/253
1	22	3933/77
2	21	3819/70
3	20	288/5
4	19	60

Each is at most 66, so Proposition 4.1 gives a K_6.

Suppose $d = 5$. Write $X = e_P(N(v), U)$ and $Y = e(P[N(v)])$. Since $N[v]$ does not span a K_6, we have $Y \le 9$. The minimum-degree inequality $X + 2Y \ge 20$ therefore gives

$$X + Y \ge 11.$$

The 18 nonneighbors of v span at most

$$70 - 5 - 11 = 54$$

edges and have independence number at most three. Proposition 3.1 again gives a K_6. $\square$

6. The 31-vertex bound

Proposition 6.1. *If H is admissible on 31 vertices and $\alpha(H) \le 5$, then*

$$e(H) \ge 97.$$

Proof. Suppose $e(H) \le 96$, choose a minimum-degree vertex v of degree $d \le 6$, and let U be its $30 - d$ nonneighbors. Then $\alpha(H[U]) \le 4$.

If $d \le 5$, Lemmas 2.4 and 2.5, with $q = 25$, give the corresponding expected edge bound below.

d	number of nonneighbors	expected edge bound
0	30	1920/29
1	29	14250/203
2	28	1550/21
3	27	1000/13
4	26	1032/13
5	25	81

Each entry is at most 81. Proposition 5.1 gives a first K_6, denoted by B_1.

If $d = 6$, write

$$X = e_H(N_H(v), U), \qquad Y = e(H[N_H(v)]).$$

The local cap on $N_H[v]$ gives $Y \le 10$, while minimum degree gives $X + 2Y \ge 30$. Hence $X + Y \ge 20$, and

$$e(H[U]) \le 96 - 6 - 20 = 70.$$

Proposition 5.2 gives B_1 in this case as well.

Apply Lemma 2.6 repeatedly. Removing B_1 leaves a 25-vertex graph Q with

$$\alpha(Q) \le 4, \qquad e(Q) \le 81.$$

Proposition 5.1 gives a second block B_2. Removing it leaves a 19-vertex graph R with

$$\alpha(R) \le 3, \qquad e(R) \le 66.$$

Proposition 4.1 gives a third block B_3. Let C be the remaining 13-vertex graph. Then

$$\alpha(C) \le 2, \qquad e(C) \le 51. \tag{16}$$

The graph C has no K_6. Suppose that $B \subseteq C$ were a K_6, and let $D = C - B$, so $|D| = 7$. If D contains a nonedge xy, each of x, y has at most one neighbor in B. Some vertex of B is nonadjacent to both, forming an independent triple with x, y, contrary to (16). If D has no nonedge, then D is a K_7, contrary to admissibility.

Put $L = \overline{C}$. By (16), the graph L is triangle-free. The absence of a K_6 in C gives $\alpha(L) \leq 5$, and hence $\Delta(L) \leq 5$. Every seven vertices span at least five edges of L, and

$$e(L) = \binom{13}{2} - e(C) \geq 78 - 51 = 27.$$

If every vertex of L had degree at most four, then $e(L) \leq 26$. Therefore L has a degree-five vertex, which has exactly seven nonneighbors. This contradicts Lemma 2.7. □

7. Proof of the main theorem

Proof of Theorem 1.1. Assume for a contradiction that every induced K_7 in a six-coloring of K_{37} uses all six colors. For each color $i \in [6]$, let G_i be its color graph. An independent seven-set in G_i would omit color i, so

$$\alpha(G_i) \leq 6. \tag{17}$$

On every seven vertices, the other five colors each occupy at least one of the 21 edges. Hence

$$e(G_i[S]) \leq 21 - 5 = 16 \qquad \text{for every } S \in \binom{V(K_{37})}{7}. \tag{18}$$

Thus each G_i is admissible.

Choose a least color graph G. The six color classes partition the $\binom{37}{2} = 666$ edges of K_{37}, so

$$e(G) \leq 111. \tag{19}$$

We claim that $\delta(G) \leq 5$. If $\delta(G) \geq 6$, then (19) forces G to be 6-regular with 111 edges. No component is K_7, by (18), and no component is an odd cycle, since every component is 6-regular. Brooks's theorem gives a proper six-coloring of each component, hence of G. One of the six independent color classes has at least seven vertices, contrary to (17).

Choose a minimum-degree vertex v of degree $d \leq 5$, and let U be its $36 - d$ nonneighbors. Then $\alpha(G[U]) \leq 5$. By Lemma 2.4,

$$e(G[U]) \leq 111 - d - \binom{d}{2}. \tag{20}$$

For $d \leq 4$, Lemma 2.5, with $q = 31$, gives the corresponding expected edge bound below.

d	number of nonneighbors	expected edge bound
0	36	1147/14
1	35	10230/119
2	34	16740/187
3	33	16275/176
4	32	1515/16

Every value is less than 96. If $d = 5$, then $|U| = 31$ and (20) gives $e(G[U]) \leq 96$ directly. In every case, some induced 31-vertex graph is admissible, has independence number at most five, and has at most 96 edges. This contradicts Proposition 6.1.

The assumed balanced coloring cannot exist. Therefore some seven vertices omit a color. □

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